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Validation Framework for Multimodal Cardiovascular AI

A Public Scientific Framework for Analytical Performance, Clinical Validity, Generalizability, Human Factors, Clinical Utility, Safety, and Lifecycle Monitoring

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Public Disclosure and Intellectual Property Boundary

This public edition is designed to establish the scientific problem, evidence expectations, validation domains, reporting principles, and research questions associated with multimodal cardiovascular artificial intelligence. It intentionally avoids implementation detail that could expose xBxBio trade secrets, patent-sensitive methods, or confidential product design.

- No xBxBio source code, executable logic, model weights, prompts, or proprietary feature engineering is disclosed.
- No production scoring formula, operating threshold, confidence boundary, escalation rule, or customer-specific configuration is disclosed.
- No detailed software architecture, internal data schema, security control design, interface specification, deployment method, or tenant configuration is disclosed.
- Illustrative validation matrices are public scientific examples and are not asserted to reproduce an xBxBio production validation protocol.
- Public modality examples are intended to clarify validation questions and should not be interpreted as product claims or regulatory indications.

Conflict of Interest Disclosure

The author is associated with xBxBio, the organization proposing the public validation framework described in this white paper. This relationship should be considered when interpreting the concepts

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and propositions presented herein. No claim of independent external validation is made unless explicitly identified and supported by documented evidence.

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Kenneth C. Bean is the sole author of this publication and is responsible for its scientific framing, analysis, interpretation, conclusions, reference selection, and approval for public release.

Disclaimer

This document presents a proposed scientific and conceptual framework for research and discussion. It does not constitute medical advice, diagnosis, treatment guidance, legal advice, regulatory authorization, clinical validation, or a substitute for professional clinical judgment. The appropriate evidence program for any actual AI system depends on its intended use, claims, users, population, setting, risk, regulatory classification, and implementation.

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Version	Status	Year	Description
0.2-P	Public Scientific Edition	2026	Initial public framework defining multimodal cardiovascular AI validation domains, evidence stages, validation matrices, lifecycle controls, and research priorities while excluding implementation-level xBxBio IP.

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1. Abstract

Multimodal cardiovascular artificial intelligence may combine electrocardiographic waveforms, cardiovascular imaging, hemodynamics, laboratory measurements, medications, clinical narratives, wearable and implantable-device data, genomic information, and longitudinal outcomes. Although combining modalities can increase informational richness, it also expands the validation problem. Performance must be established not only for individual inputs and model outputs, but also for modality alignment, temporal coherence, missingness, cross-modal conflict, data provenance, workflow interaction, external generalizability, and post-deployment change.

This white paper proposes a public scientific validation framework for multimodal cardiovascular AI. The framework separates analytical performance, clinical validity, human-factors performance, clinical utility, software and cybersecurity assurance, and lifecycle monitoring. It treats intended use and claims as the organizing center of validation and emphasizes that the appropriate evidence program must be proportional to clinical consequence, uncertainty, population, care setting, and the role assigned to the human user.

The framework distinguishes internal testing from external testing, uses reference-standard language rather than assuming absolute ground truth, and proposes validation domains for data quality, modality-specific behavior, fusion logic, temporal robustness, calibration, subgroup performance, out-of-distribution behavior, abstention and escalation, human-AI interaction, workflow effects, and real-world monitoring. It aligns conceptually with current regulatory, standards, and reporting expectations, including total-product-lifecycle risk management, Good Machine Learning Practice, medical-device risk management, software lifecycle controls, usability engineering, clinical investigation principles, and reporting guidance for prediction, diagnostic-accuracy, imaging-AI, and early clinical evaluation studies.[1-21]

This is a proposed and testable xBxBio public framework. It does not establish that any particular AI system is safe, effective, clinically useful, generalizable, equitable, or regulatorily authorized. The central proposition is that multimodal cardiovascular AI requires a validation system that evaluates the complete evidence pathway - from source data through integration, inference, human interpretation, clinical action, and lifecycle monitoring - rather than relying on a single aggregate performance metric.

2. Introduction

Artificial intelligence in cardiovascular medicine has expanded from narrow signal-classification tasks toward systems that may integrate multiple data types and support diagnosis, prognosis, triage, monitoring, workflow prioritization, or treatment-related decisions. Reviews and scientific statements describe opportunities across imaging, electrocardiography, electronic health records, wearables, genetics, and remote monitoring, while also emphasizing unresolved questions concerning evidence, bias, implementation, reliability, and clinical benefit.[22-24]

Multimodality does not merely add more predictors. It creates interaction effects and failure modes that are absent in a single-modality model. A clinically plausible output can result from the wrong patient linkage, mismatched time windows, correlated data leakage, a dominant low-quality modality, an inappropriate fallback rule, or a fusion process that masks disagreement. A system can perform well on average while becoming unreliable when one input is missing, delayed, technically degraded, acquired with a different device, or collected after a major intervention.

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Validation therefore must be designed as a system of evidence. The validation target is not only an algorithm. It includes the data pipeline, preprocessing, modality-specific components, multimodal integration, user interface, intended workflow, operating environment, human-AI team, software lifecycle, cybersecurity posture, and post-deployment monitoring process.

Current FDA and IMDRF materials emphasize total-product-lifecycle risk management, representative datasets, separation of training and test activities, performance of the human-AI team, transparency, monitoring, and controlled change.[1-5] Scientific reporting frameworks similarly stress intended use, transparent data description, external testing, reference standards, risk of bias, human factors, and prospective evaluation.[15-21] This paper organizes those ideas into a cardiovascular-specific public validation architecture.

2.1 Central Research Question

What evidence is required to establish that a multimodal cardiovascular AI system performs reliably, safely, equitably, and usefully across its intended data sources, patient populations, clinical settings, human workflows, and lifecycle changes?

2.2 Public Framework Objectives

- Define validation domains that extend beyond a single model-performance number.
- Make intended use, clinical claims, users, workflow, and consequence explicit before selecting validation methods.
- Separate data, analytical, clinical, human-factors, utility, software, cybersecurity, and lifecycle evidence.
- Address multimodal and longitudinal failure modes, including missingness, asynchronous evidence, and cross-modal conflict.
- Support transparent reporting and external review without exposing proprietary implementation methods.
- Provide a research agenda for prospective evaluation of multimodal cardiovascular AI.

3. Scope, Definitions, and Evidence Boundary

3.1 Multimodal Cardiovascular AI

For this paper, multimodal cardiovascular AI means an AI-enabled system that uses, compares, or coordinates two or more clinically relevant information modalities to generate or support a cardiovascular output. Modalities may include waveforms, images, structured observations, text, laboratory data, medication history, hemodynamics, wearable or implantable-device streams, genomics, and patient-reported information.

3.2 Validation

Validation is used here as an umbrella term for the evidence activities needed to establish fitness for an intended use. Because the word can be ambiguous, individual studies should state whether they assess data quality, internal testing, external testing, analytical performance, diagnostic accuracy, prediction performance, usability, clinical utility, or another defined objective. CLAIM 2024 specifically encourages precise terms such as internal testing and external testing rather than using validation without qualification.[18]

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3.3 Reference Standard

A reference standard is the method used to establish the comparison target for a study. It may be based on expert adjudication, invasive measurement, longitudinal outcome, pathology, imaging, consensus criteria, or another fit-for-purpose source. The reference standard can itself be uncertain or biased; therefore its construction and limitations require validation and transparent reporting.[17-18]

3.4 Evidence Boundary

- This framework does not prescribe a universal regulatory pathway.
- It does not assume that all multimodal AI functions are medical devices.
- It does not specify proprietary model architecture, fusion methods, or operating thresholds.
- It does not treat statistical significance as equivalent to clinical utility.
- It does not infer clinical benefit from retrospective performance alone.
- It does not replace standards, regulatory guidance, clinical-investigation requirements, or independent expert review.

4. Intended Use, Claims, and Validation Strategy

Validation begins with a precise intended use. The same algorithm can require materially different evidence depending on whether it is used for research, documentation support, screening, diagnostic assistance, risk prediction, triage, monitoring, treatment recommendation, or autonomous action. Intended users, patient population, care setting, input sources, output format, action pathway, contraindications, and expected clinical consequences should be defined before the validation plan is finalized.

Claims determine evidence burden. A claim that a system reproduces a measurement requires evidence different from a claim that it improves diagnostic accuracy, changes clinician behavior, shortens time to treatment, reduces hospitalization, or improves patient outcomes. Validation should therefore map each claim to a defined evidence endpoint and study design.

For applicable device software, FDA materials emphasize documentation that supports safety and effectiveness throughout the total product lifecycle, while the 2026 CDS guidance distinguishes certain non-device CDS functions from device software based on statutory criteria and the ability of health professionals to independently review the basis for recommendations.[1,5] This paper does not classify any specific xBxBio function; it emphasizes that classification and evidence planning depend on actual intended use and functionality.

4.1 Minimum Intended-Use Specification

Specification element	Validation question
Clinical purpose	What cardiovascular question or workflow need is addressed?
Target population	Which patients, subgroups, disease states, and exclusions apply?
Care setting	Ambulatory, inpatient, emergency, imaging laboratory, home, remote monitoring, research, or other setting.
Intended user	Cardiologist, imaging specialist, nurse, primary-care

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Specification element	Validation question
	clinician, patient, caregiver, researcher, or operational user.
Input modalities	Required and optional data sources, acquisition conditions, timing, and quality requirements.
Output and action	Classification, measurement, risk estimate, recommendation, prioritization, alert, summary, or evidence request.
Human role	Review, confirmation, override, escalation, independent interpretation, or other responsibility.
Clinical consequence	Potential harm from false positive, false negative, delay, inappropriate action, or missed escalation.

5. Validation as a Layered Evidence System

A multimodal cardiovascular AI system should be validated across distinct but connected layers. Evidence at one layer cannot automatically substitute for evidence at another. Strong analytical performance does not establish workflow usability; strong usability does not establish clinical validity; strong clinical validity does not establish improved outcomes; and predeployment performance does not establish sustained post-deployment reliability.

Evidence layer	Core validation purpose
Layer 1 - Data integrity	Identity, provenance, completeness, representativeness, acquisition quality, temporal metadata, and transformation lineage.
Layer 2 - Modality analytics	Performance of each modality-specific input, preprocessing step, feature, measurement, segmentation, or submodel.
Layer 3 - Multimodal integration	Alignment, fusion, weighting, missing-modality behavior, disagreement handling, and interaction effects.
Layer 4 - Clinical validity	Association of the system output with the clinical condition, event, or outcome defined by the intended use.
Layer 5 - Human factors	Comprehension, use error, automation bias, override behavior, alert burden, and performance of the human-AI team.
Layer 6 - Clinical utility	Effect on decisions, workflow, resource use, safety, patient outcomes, or other prespecified benefits and harms.
Layer 7 - Lifecycle assurance	Software integrity, cybersecurity, change control, drift monitoring, incident response, and continued performance.

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5.1 Traceability Across Layers

The validation file should trace intended-use claims to hazards, requirements, datasets, test cases, acceptance criteria, results, residual risks, labeling, monitoring metrics, and change-control decisions. Traceability is particularly important when a final multimodal output depends on multiple upstream components that may change independently.

6. Data and Dataset Validation

Data validation establishes whether the evidence used to develop and evaluate the system is sufficiently accurate, relevant, representative, independent, and traceable for the intended use. It is not limited to checking whether files can be read. It includes patient identity, encounter linkage, modality acquisition, timestamps, units, annotation, missingness, preprocessing, exclusions, and the relationship between the dataset and the target population.

Good Machine Learning Practice emphasizes representative datasets, independence between training and test sets, fit-for-purpose reference datasets, attention to the human-AI team, and monitoring after deployment.[2] TRIPOD+AI, PROBAST+AI, STARD-AI, and CLAIM provide complementary reporting and bias-assessment expectations for prediction, diagnostic-accuracy, and imaging-AI studies.[15-18]

6.1 Dataset Validation Questions

- Are patients, encounters, and observations linked correctly across modalities?
- Do date and time fields distinguish acquisition, observation, documentation, result availability, and model inference?
- Are training, tuning, internal testing, and external testing partitions independent at the patient and site level where appropriate?
- Could data leakage occur through repeated examinations, derived labels, post-outcome documentation, site-specific artifacts, or duplicated records?
- Are demographic, clinical, device, geographic, and care-setting distributions consistent with the intended population?
- Are exclusion criteria transparent, clinically justified, and evaluated for selection bias?
- Are missingness patterns characterized rather than treated as random by default?
- Are source-system transformations, unit conversions, normalization, de-identification, and manual corrections documented?
- Can each validation result be reconstructed from versioned data and code?

6.2 Dataset Shift Taxonomy

Shift type	Illustrative validation concern
Population shift	Change in patient characteristics, disease prevalence, severity, referral patterns, or treatment patterns.
Acquisition shift	Different device, sensor, imaging protocol, signal sampling, laboratory platform, or recording environment.
Workflow shift	Different ordering behavior, documentation practice, staffing, user role, or timing of care.
Label shift	Changed prevalence or relationship between labels

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Shift type	Illustrative validation concern
	and underlying disease state.
Concept shift	Changed relationship between input evidence and outcome, including treatment or guideline changes.
Temporal shift	Performance changes across calendar time, season, intervention era, or evolving clinical practice.

7. Modality-Specific Analytical Validation

Each modality requires evidence appropriate to its acquisition process, technical characteristics, clinical meaning, and failure modes. Multimodal performance cannot compensate for an unvalidated or unreliable component when that component can materially influence the final output.

Modality	Public analytical-validation considerations
ECG and waveforms	Sampling rate, lead configuration, filtering, noise, rhythm prevalence, interval measurement, arrhythmia reference standard, device variability, and signal-quality rejection.
Echocardiography	View recognition, segmentation, measurement reproducibility, reader variability, machine and protocol variation, image quality, frame selection, and physiologic context.
CT / CMR / nuclear imaging	Acquisition protocol, reconstruction, contrast timing, sequence selection, segmentation, quantitative mapping, vendor/site variability, and artifact sensitivity.
Hemodynamics	Calibration, zeroing, transducer position, waveform quality, beat selection, physiologic state, invasive-reference uncertainty, and unit consistency.
Laboratory data	Assay method, reference ranges, units, specimen timing, hemolysis or interference, platform differences, and treatment-related change.
Wearables and implantables	Device identity, wear time, sensor contact, missingness, firmware, activity context, signal quality, population adherence, and confirmation pathway.
Clinical text	Documentation timing, authorship, copy-forward, negation, uncertainty, note type, leakage from future events, and terminology variation.
Genomics and molecular data	Variant calling, ancestry representation, assay quality, interpretation framework, penetrance uncertainty, and evolving knowledge.

7.1 Acceptance Criteria Must Be Clinically Linked

Acceptance criteria should reflect the clinical role of the measurement or classification. A small measurement error may be acceptable when it does not change interpretation, but clinically

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consequential when it crosses a treatment, severity, or referral boundary. Therefore validation should examine both continuous error and decision-relevant error.

8. Multimodal Integration and Fusion Validation

Multimodal integration is a distinct validation target. It includes the rules, learned representations, or workflow logic by which modalities are aligned and combined. The final system can fail even when every modality-specific component performs within specification.

Validation should test whether the system uses the intended patient, time window, clinical event, and modality version. It should also examine dominance, redundancy, contradiction, missingness, and interaction. A high-volume modality should not overwhelm a lower-volume but clinically decisive modality merely because it contributes more features.

Because proprietary fusion methods need not be publicly disclosed, public reporting can focus on validation behavior: which modalities are required, what happens when one is absent, how conflicting evidence is surfaced, how temporally incompatible inputs are handled, and whether performance changes under clinically realistic combinations.

8.1 Integration Test Families

- Complete-input testing: all required and optional modalities are present and within quality limits.
- Missing-modality testing: one or more modalities are absent, delayed, or unavailable.
- Degraded-modality testing: a modality is present but noisy, low quality, incomplete, or uncertain.
- Conflict testing: modalities support materially different interpretations.
- Asynchronous testing: modalities represent different clinical states or acquisition times.
- Redundancy testing: duplicated or highly correlated evidence is introduced.
- Dominance testing: one modality is extreme, highly confident, or disproportionately represented.
- Mismatch testing: data from the wrong patient, encounter, device, or time window are introduced.
- Fallback testing: the system enters a reduced-input, abstention, evidence-request, or escalation state.

9. Temporal and Longitudinal Validation

Cardiovascular evidence changes across time and in response to interventions. A model may combine data that are individually correct but collectively describe different patient states. Temporal validation examines whether the system recognizes clinically material events and applies appropriate time relationships.

The validation dataset should distinguish at least observation time, acquisition time, result availability, documentation time, intervention time, and inference time when these differences can affect the intended use. It should also test whether evidence remains valid after medication changes, procedures, hospitalization, acute illness, rhythm conversion, device implantation, or other state-changing events.

Longitudinal performance should be evaluated for repeated observations from the same patient, correlated errors, changing prevalence, feedback from prior AI outputs, and the effect of clinical actions on future data. Standard random splits may overstate performance when repeated patient data cross partitions.

9.1 Temporal Stress Scenarios

- Pre-intervention imaging combined with post-intervention laboratory or hemodynamic data.

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- An ECG acquired during an acute rhythm event combined with baseline imaging from stable sinus rhythm.
- Wearable trends that precede formal diagnosis but are documented after the diagnosis.
- Laboratory values that are recent in clock time but obsolete after a major medication or volume-status change.
- Delayed external records that arrive after a clinical decision has already occurred.
- A monitoring model that is retrained on data influenced by earlier model-triggered interventions.

10. Reference Standards, Adjudication, and Label Quality

Reference-standard quality can limit the maximum credibility of a validation study. Cardiovascular labels may be based on expert interpretation, consensus criteria, invasive measurement, longitudinal outcomes, pathology, or combinations of evidence. Each approach has uncertainty, variability, and potential bias.

Adjudication should be designed independently of model output where feasible. Reviewers should receive standardized instructions, relevant clinical context, and predefined disagreement procedures. Blinding, reader number, expertise, inter-rater agreement, and adjudication sequence should be reported.

For composite or longitudinal reference standards, the study should specify which evidence is available at each time point and avoid using future information that would not have been available to the model or clinician at the intended decision time.

10.1 Reference-Standard Design Options

Reference-standard approach	Validation consideration
Single expert	Efficient but vulnerable to individual variability and bias; requires expertise justification.
Multiple independent experts	Supports agreement assessment and reduces individual dependence.
Consensus panel	Useful for complex cases but can obscure initial disagreement unless documented.
Objective measurement	Can reduce interpretive variability but may have technical error or imperfect clinical relevance.
Clinical outcome	Directly relevant for prognosis but may be affected by treatment, follow-up, competing risk, and ascertainment.
Composite standard	Can integrate multiple sources but requires prespecified logic and sensitivity analysis.

11. Statistical Performance, Calibration, and Uncertainty

Metric selection should match the intended use, outcome prevalence, decision consequence, and study design. A single discrimination metric is rarely sufficient. Validation may require sensitivity, specificity, predictive values, likelihood ratios, calibration, decision-curve analysis, measurement error, agreement, time-to-event performance, and clinically relevant operating points.

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Confidence intervals and uncertainty estimates should accompany primary metrics. Sample-size planning should consider the number of events, class imbalance, clustering by patient or site, repeated measures, subgroup analyses, and the need to estimate performance at clinically relevant thresholds.

Calibration should be evaluated in the intended population and care setting. A model can discriminate well while producing probabilities that are systematically too high or too low. Recalibration can alter performance and should be treated as a controlled model change rather than an undocumented local adjustment.

Uncertainty has multiple sources: data quality, label uncertainty, sampling variation, model uncertainty, distribution shift, missingness, and human interpretation. Validation should identify which uncertainty is measured, how it is communicated, and how it affects permitted clinical behavior.

11.1 Minimum Metric Categories

Metric category	Examples
Discrimination	AUROC, precision-recall, C-index, or other task-appropriate measures.
Operating-point performance	Sensitivity, specificity, predictive values, false-alert rate, missed-event rate.
Calibration	Calibration plot, calibration-in-the-large, slope, Brier score, or fit-for-purpose alternatives.
Agreement and measurement	Bias, limits of agreement, repeatability, reproducibility, intraclass correlation, clinically meaningful error.
Utility	Net benefit, decision impact, time-to-action, resource use, or prespecified outcome effect.
Uncertainty	Confidence intervals, bootstrap variation, posterior intervals, or other justified uncertainty estimates.

12. External Testing and Generalizability

External testing should evaluate performance in data that are meaningfully independent of model development. Independence can involve different institutions, geographic regions, time periods, devices, acquisition protocols, populations, workflows, or clinical prevalence. Merely holding out a random subset from the same source does not establish external generalizability.

The external-testing strategy should be selected based on intended use. A multi-hospital inpatient model may require site-level external testing; a wearable model may require device and home-environment variation; an imaging model may require vendor and protocol diversity; and a longitudinal model may require testing across changing treatment eras.

Generalizability is not binary. Performance can be adequate in some subgroups or settings and inadequate in others. Validation should identify the supported operating domain and define what evidence is needed before expansion.

12.1 External-Test Dimensions

- Institution and health-system independence.

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- Geographic and socioeconomic diversity.
- Device, vendor, protocol, and software-version diversity.
- Prospective temporal separation from model development.
- Care-setting diversity and workflow realism.
- Disease prevalence, severity, comorbidity, and treatment-pattern variation.
- Language, documentation, coding, and interoperability variation.
- Performance under local missingness and data-availability patterns.

13. Robustness, Missingness, Noise, and Conflict

Robustness testing evaluates whether the system behaves acceptably under foreseeable perturbations. For multimodal cardiovascular AI, this includes both natural variation and engineered failure injection. Robustness does not mean performance remains unchanged; it means degradation, abstention, evidence request, or escalation occurs in a controlled and clinically appropriate manner.

Robustness domain	Illustrative challenge
Signal degradation	Noise, motion, missing leads, image artifact, poor sensor contact, truncated waveform, corrupted file.
Missing data	Absent modality, missing segment, unavailable external record, delayed laboratory result, incomplete follow-up.
Unit and scale errors	Incorrect unit, decimal shift, normalization failure, device-specific scaling, reference-range mismatch.
Identity and linkage errors	Wrong patient, duplicate record, encounter mismatch, device attribution error.
Semantic conflict	Contradictory codes, notes, measurements, diagnoses, or modality interpretations.
Out-of-domain input	Unsupported device, pediatric patient in adult-only model, rare condition, unvalidated protocol.
Adversarial or cyber event	Manipulated input, corrupted model artifact, unauthorized configuration change, unavailable dependency.

13.1 Failure Behavior Is a Validation Endpoint

A safe system may be expected to refuse an output, request additional evidence, qualify uncertainty, or escalate to a clinician. These behaviors require testable acceptance criteria. Validation should measure missed abstention, unnecessary abstention, inappropriate fallback, delayed escalation, and user misunderstanding of system silence.

14. Human Factors and Clinical Workflow Validation

The performance of a clinical AI system depends on the human-AI team. FDA transparency principles emphasize the information needs of users and the performance of the human-AI team, while IEC 62366-1 provides a usability-engineering process related to safety.[3,12] DECIDE-AI emphasizes early evaluation in live clinical settings, including human factors and implementation context.[19]

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Human-factors validation should evaluate whether users understand the output, limitations, evidence basis, uncertainty, required action, and override pathway. It should also test use error, automation bias, alert fatigue, confirmation bias, anchoring, over-reliance, under-reliance, and the effect of time pressure.

Workflow validation should compare the AI-supported process with current practice. It should examine who receives the output, when it appears, whether relevant evidence is accessible, how responsibility is assigned, what happens when the output conflicts with clinician judgment, and how unresolved alerts are managed.

14.1 Human-Factors Evidence Stages

Stage	Primary purpose
Formative evaluation	Identify comprehension, navigation, terminology, display, and workflow problems during design.
Simulated-use testing	Test realistic tasks, time pressure, interruptions, error recovery, and uncommon scenarios.
Summative usability validation	Demonstrate that intended users can perform critical tasks safely and effectively under defined conditions.
Early live clinical evaluation	Observe real-world interaction, workflow fit, learning curve, and unintended consequences.
Post-deployment monitoring	Track overrides, alert response, use error, workarounds, complaints, and evolving user behavior.

15. Clinical Utility, Effectiveness, and Outcome Evidence

Clinical validity establishes that an output relates to the intended clinical condition or outcome. Clinical utility asks whether using the system improves decisions, workflow, safety, resource use, patient experience, or health outcomes relative to an appropriate comparator.

Retrospective model performance does not establish clinical utility. A system can be accurate but unused, misunderstood, delayed, duplicative, or disruptive. It can also shift resources or testing without improving outcomes. Utility studies should prespecify the mechanism by which the AI is expected to produce benefit and measure potential harms.

Study designs may include prospective observational studies, silent deployment, stepped-wedge designs, randomized trials, cluster trials, diagnostic-impact studies, workflow studies, or pragmatic evaluations. SPIRIT-AI and CONSORT-AI support transparent protocol and trial reporting; DECIDE-AI supports early live clinical evaluation.[19-21] ISO 14155 and ICH E6(R3) provide broader clinical-investigation and Good Clinical Practice principles.[13-14]

15.1 Utility Endpoint Categories

- Diagnostic or prognostic decision accuracy.
- Time to clinically appropriate action.
- Changes in testing, referral, treatment, or monitoring.
- Hospitalization, complications, mortality, symptoms, function, or quality of life where appropriate.
- Clinician workload, alert burden, documentation time, and cognitive demand.

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- Resource use, cost, equity, and access.
- Unintended consequences, including overdiagnosis, overtreatment, delay, false reassurance, and deskilling.

16. Fairness, Subgroup Performance, and Access

Aggregate performance can conceal clinically important subgroup differences. Validation should identify subgroups relevant to the intended use, including age, sex, race and ethnicity where scientifically and ethically appropriate, socioeconomic context, geography, disease phenotype, comorbidity, device type, site, language, and data availability.

Fairness assessment should consider both model error and pathway effects. A system can have similar sensitivity across groups but produce unequal benefit if access to confirmatory testing, specialists, devices, or follow-up differs. Missingness and modality availability can also create unequal abstention or evidence-request burden.

Subgroup analyses require adequate sample size and careful interpretation. Small samples can produce unstable estimates, while overreliance on broad demographic categories can conceal clinically meaningful heterogeneity. The validation plan should distinguish prespecified analyses from exploratory analyses and avoid implying equivalence from nonsignificant differences.

16.1 Equity-Oriented Questions

- Who is underrepresented in development and external-testing datasets?
- Do acquisition quality or missing-modality rates differ across populations or care settings?
- Are false negatives, false positives, abstentions, or escalations distributed unequally?
- Does the system require technology or specialist access that is not uniformly available?
- Can local recalibration or workflow adaptation create new disparities?
- Are patients and clinicians able to understand how data are used and challenge incorrect outputs?

17. Transparency, Provenance, and Auditability

Transparency should be designed for the intended audience. Developers, validators, regulators, clinicians, patients, administrators, and incident investigators require different information. FDA transparency principles frame transparency as communicating appropriate information about intended use, development, performance, and logic to relevant audiences.[3]

For validation, provenance means the ability to identify source data, acquisition method, transformation history, model and rule version, inference time, configuration, and user-visible output. Auditability means that a validation result or clinical inference can be reconstructed sufficiently to investigate performance, bias, error, or change.

Public transparency does not require disclosure of trade secrets. A public report can describe intended use, datasets, reference standards, performance, limitations, validation design, monitoring, and known failure modes without publishing proprietary source code, thresholds, or architecture.

17.1 Minimum Inference Audit Record

- Patient and encounter identifiers or privacy-preserving equivalents.
- Source modalities, acquisition times, and quality state.
- Preprocessing and data-transformation versions.
- Model, rule, and configuration versions.

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- Output, uncertainty, abstention, or escalation state.
- User-visible explanation and limitations.
- Clinician response, override, or disposition when applicable.
- Outcome and follow-up information when available for monitoring.

18. Safety Engineering, Risk Management, and Failure Analysis

Validation should be integrated with risk management rather than treated as a separate statistical exercise. ISO 14971 specifies a lifecycle process for identifying hazards, estimating and evaluating risk, implementing controls, and monitoring effectiveness; FDA AI guidance similarly emphasizes total-product-lifecycle risk management.[1,10]

Potential harms can arise from incorrect output, delayed output, unavailable output, misleading confidence, data mismatch, poor user interpretation, workflow displacement, cybersecurity failure, or inappropriate reliance. The validation plan should test risk controls and verify that residual risks remain acceptable for the intended use.

Failure analysis should examine not only average errors but also clinically meaningful error clusters. Incorrect cases should be characterized by modality, disease phenotype, subgroup, site, device, acquisition quality, timing, and workflow response. Root-cause investigation should distinguish data, model, integration, interface, user, and operational causes.

18.1 Public Hazard Categories

Hazard category	Illustrative manifestation
Incorrect clinical inference	False positive, false negative, wrong measurement, wrong prioritization, or misleading summary.
Temporal error	Stale evidence, future leakage, missed intervention, or incorrect sequence.
Identity/provenance error	Wrong patient, encounter, source, device, or transformation.
Human-use error	Misinterpretation, missed alert, inappropriate override, over-reliance, or under-reliance.
Availability failure	System downtime, dependency failure, delayed result, or failed fallback.
Cybersecurity failure	Unauthorized access, manipulation, integrity loss, denial of service, or unsafe update.
Equity harm	Disproportionate error, abstention, access burden, or resource displacement.

19. Software Lifecycle, Cybersecurity, and Change Control

Multimodal AI depends on software, interfaces, models, data transformations, external services, and infrastructure. Validation should therefore include software lifecycle assurance and cybersecurity, not only model testing. IEC 62304 defines medical-device software lifecycle processes; FDA cybersecurity

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guidance addresses secure design, documentation, and resilience for devices with cybersecurity risk. [6,11]

Changes to models, data pipelines, modalities, interfaces, devices, operating systems, reference standards, or user workflows can alter performance. A change-impact assessment should identify which evidence remains valid and which tests must be repeated. The FDA PCCP guidance provides recommendations for planned AI-enabled device modifications and associated development, validation, implementation, and impact assessment for applicable devices.[4]

The public framework does not prescribe a particular update mechanism. It requires that updates be versioned, risk assessed, tested, approved, monitored, and traceable. Silent or uncontrolled changes undermine the interpretability of both predeployment and post-deployment evidence.

19.1 Change Categories Requiring Validation Review

- New or retrained model version.
- New modality, device, vendor, protocol, or site.
- Changed preprocessing, normalization, feature extraction, or fusion behavior.
- Changed threshold, labeling, intended use, population, user, or workflow.
- Changed software dependency, hosting, interface, or cybersecurity control.
- Changed reference standard or clinical guideline.
- Changed monitoring metric, acceptance criterion, or fallback behavior.

20. Post-Deployment Monitoring and Real-World Performance

Post-deployment monitoring should evaluate continued performance in the real operating environment. Monitoring must distinguish changes in patient population, data availability, devices, workflow, user behavior, clinical practice, and model performance. It should also identify safety events that are not visible in aggregate metrics.

A monitoring plan should define metrics, baselines, alert limits, review frequency, subgroup stratification, investigation responsibility, escalation, corrective action, and criteria for suspension or rollback. Monitoring should include both technical outcomes and clinical workflow outcomes.

Real-world performance data can support continued assurance but can also be biased by selective follow-up, changing treatment, feedback loops, incomplete outcomes, and unmeasured confounding. Monitoring results should therefore be interpreted within a prespecified causal and operational framework rather than treated as self-validating evidence.

20.1 Suggested Monitoring Metrics

Monitoring domain	Example measures
Input health	Missingness, quality failures, modality availability, source changes, latency, unit errors.
Model performance	Calibration, discrimination, operating-point metrics, measurement error, drift.
State behavior	Abstention, evidence request, escalation, fallback, unavailable output.
Human response	Acknowledgment, override, time to action, use error,

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Monitoring domain	Example measures
	alert fatigue, workarounds.
Clinical process	Testing, referral, treatment, monitoring, delay, resource use.
Safety and quality	Complaints, near misses, incidents, CAPA signals, cybersecurity events.
Equity	Subgroup error, access, missingness, abstention burden, and workflow impact.

21. Proposed Public Multimodal Validation Matrix

The following matrix is a public planning tool. It is intentionally high level and does not disclose xBxBio implementation methods or proprietary acceptance thresholds.

Validation domain	Public evidence objective	Illustrative evidence method
Data identity and provenance	All modalities link to correct patient, encounter, source, and time.	Mismatch, duplication, transformation, unit, and source-change tests.
Modality analytics	Each input component performs for intended devices, protocols, and quality ranges.	Internal and external testing with modality-specific reference standards.
Multimodal integration	Fusion behaves appropriately under complete, missing, degraded, conflicting, and asynchronous inputs.	Scenario testing, failure injection, ablation, sensitivity, and fallback evaluation.
Clinical validity	Output relates to intended condition, event, or outcome.	Diagnostic, prognostic, measurement, or monitoring study aligned to claims.
Human factors	Users understand and act on output safely.	Formative, simulated-use, summative, and early live clinical evaluation.
Clinical utility	Use produces prespecified beneficial effects with acceptable harms.	Prospective impact or outcome evaluation against comparator.
Generalizability	Performance remains acceptable across intended sites, devices, populations, and time.	External testing and subgroup analysis.
Lifecycle assurance	Performance remains controlled through updates, cyber events, drift, and workflow change.	Change control, regression testing, monitoring, incident response, and rollback.

22. Evidence Maturity and Release Gates

Evidence should mature through defined gates. A system should not advance merely because a model metric improved. Advancement should depend on closure of the evidence activities relevant to its intended use and risk.

Evidence gate	Minimum public meaning
Gate 0 - Concept	Clinical need, intended use, users, modalities, claims,

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Evidence gate	Minimum public meaning
	and risks are defined.
Gate 1 - Data readiness	Data provenance, representativeness, reference standards, partitions, and quality controls are established.
Gate 2 - Analytical evidence	Modality and integration performance meet prespecified criteria under controlled testing.
Gate 3 - External evidence	Performance is demonstrated in independent populations, sites, devices, or time periods.
Gate 4 - Human and workflow evidence	Usability, comprehension, use safety, and workflow behavior are acceptable.
Gate 5 - Clinical utility evidence	The AI-supported pathway demonstrates fit-for-purpose clinical or process benefit with characterized harms.
Gate 6 - Deployment readiness	Software, cybersecurity, monitoring, change control, labeling, training, and incident response are complete.
Gate 7 - Continued assurance	Real-world performance remains within control and changes are managed through the lifecycle.

22.1 Gate Decisions

- Advance: evidence is sufficient for the next controlled stage.
- Advance with conditions: defined limitations, monitoring, or additional evidence are required.
- Hold: material evidence gaps remain.
- Narrow: intended use, claims, population, modality, or workflow must be reduced.
- Repeat: evidence is invalidated by change, bias, protocol deviation, or failed acceptance criteria.
- Stop: residual risk or lack of benefit does not justify continued development or deployment.

23. Research Agenda

The framework is proposed as a research program. Priority questions include:

- Which validation failures are unique to multimodal cardiovascular AI compared with single-modality systems?
- How should reference standards be constructed when modalities disagree and no definitive test exists?
- What study designs best evaluate cross-modal fusion without exposing proprietary architecture?
- How should temporal validity and intervention-related state change be incorporated into external testing?
- What metrics best measure safe degradation, abstention, evidence request, and escalation?
- How can external testing be designed when modality availability differs substantially across sites?
- What evidence is sufficient to support local adaptation or recalibration?
- How should subgroup fairness be evaluated when sample sizes are small and modalities are unequally available?
- Can state and provenance telemetry detect upstream failure before clinical harm occurs?

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- How do clinicians interpret multimodal explanations and resolve cross-modal conflict?
- Which prospective designs most efficiently establish clinical utility while protecting patients?
- How should post-deployment evidence be linked to change control, incident investigation, and continued authorization?

23.1 Suggested Study Program

Study phase	Public objective
Phase 1 - Specification	Define intended use, claims, risks, modalities, reference standards, acceptance criteria, and reporting plan.
Phase 2 - Retrospective analytical testing	Evaluate data, modality, fusion, temporal, robustness, calibration, and subgroup performance.
Phase 3 - Independent external testing	Assess generalizability across sites, devices, protocols, populations, and time.
Phase 4 - Human-factors and silent deployment	Evaluate usability, workflow, alert behavior, and operational performance without influencing care.
Phase 5 - Early live clinical evaluation	Assess real clinical use, safety, human-AI interaction, and iterative implementation.
Phase 6 - Comparative utility or outcome study	Test prespecified clinical, workflow, economic, or patient-centered benefits and harms.
Phase 7 - Lifecycle monitoring	Evaluate drift, subgroup behavior, change control, incident signals, and continued performance.

24. Limitations

- This is a proposed xBxBio public framework and has not been independently validated as a universal validation standard.
- The appropriate evidence program depends on intended use, claims, population, setting, risk, regulatory classification, and implementation.
- The framework is broad and cannot replace modality-specific, disease-specific, statistical, clinical, regulatory, or standards expertise.
- Public reporting cannot fully demonstrate proprietary implementation behavior, and confidentiality limits can complicate reproducibility.
- Many multimodal validation questions lack agreed metrics or reference datasets.
- External testing may not predict performance in every future site, population, device, or workflow.
- Clinical utility and patient benefit cannot be inferred from retrospective performance, calibration, or usability alone.
- Rapidly changing regulation, standards, and AI technology require source re-verification immediately before final public release.

25. Conclusions

Multimodal cardiovascular AI requires more than a larger dataset and a stronger aggregate performance metric. It requires validation of the complete evidence pathway: source data, modality-specific analytics, cross-modal integration, temporal context, reference standards, statistical

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performance, generalizability, human interpretation, clinical action, software integrity, cybersecurity, and lifecycle monitoring.

The framework proposed here organizes validation around intended use and clinical consequence. It distinguishes analytical performance, clinical validity, human factors, clinical utility, and continued assurance, while treating missingness, cross-modal conflict, uncertainty, subgroup performance, and safe failure behavior as first-class validation targets.

The central proposition is testable: a layered, traceable, consequence-sensitive validation program should provide more credible evidence for multimodal cardiovascular AI than reliance on a single retrospective performance study. Whether this framework improves safety, clinical utility, equity, or regulatory efficiency requires independent review and prospective research.

xBxBio advances this public framework to support scientific discussion while preserving the boundary between public validation principles and confidential implementation-level intellectual property.

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Appendix A - Public Validation Planning Checklist

Planning domain	Minimum public checklist item
Intended use	Clinical purpose, users, population, setting, modalities, output, action, and limitations are defined.
Claims map	Each public or regulatory claim is linked to a prespecified endpoint and evidence source.
Risk map	Hazards, clinical consequence, risk controls, and residual-risk criteria are documented.
Dataset plan	Sources, partitions, representativeness, missingness, leakage controls, and provenance are defined.
Reference standard	Method, readers, blinding, disagreement, uncertainty, and limitations are defined.
Modality testing	Each modality has device/protocol/quality-specific test coverage and acceptance criteria.
Integration testing	Missing, degraded, conflicting, asynchronous, and mismatched modality scenarios are tested.
Statistical plan	Primary metrics, sample size, uncertainty, multiplicity, clustering, and calibration are prespecified.
External testing	Independent sites, time periods, devices, populations, and workflows are included as needed.
Human factors	Critical tasks, use errors, comprehension, overrides, alert burden, and workflow fit are evaluated.
Clinical utility	Comparator, mechanism of benefit, outcomes, harms, and prospective design are defined.
Fairness	Subgroups, access, missingness, abstention, and pathway effects are evaluated.
Software/cyber	Lifecycle, configuration, cybersecurity, dependencies, and update controls are verified.
Monitoring	Metrics, limits, review cadence, escalation, CAPA, rollback, and revalidation criteria are defined.
Public reporting	Methods, results, limitations, version, date, and IP boundary are reported transparently.

Appendix B - Multimodal Failure-Injection Matrix

Failure domain	Public test example	Expected validation behavior
Identity	Replace one modality with another patient or encounter.	Detect mismatch, reject, abstain, or escalate according to intended use.
Time	Shift one modality across a major intervention or acute event.	Recognize temporal incompatibility and prevent unsupported fusion.

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Failure domain	Public test example	Expected validation behavior
Signal quality	Add noise, artifact, missing leads, low image quality, or sensor dropout.	Degrade safely; communicate quality; avoid false confidence.
Missing modality	Remove required or optional input.	Follow prespecified missing-input behavior.
Conflict	Create clinically plausible but discordant modalities.	Surface disagreement; avoid opaque averaging; support review.
Units	Introduce unit mismatch, decimal shift, or reference-range error.	Detect plausibility or data-integrity failure.
Device/site	Use unvalidated device, protocol, or site distribution.	Identify out-of-domain condition or demonstrate controlled performance.
Label	Introduce uncertain, inconsistent, or biased reference labels.	Quantify sensitivity to reference-standard uncertainty.
Cyber/integrity	Modify input, model artifact, configuration, or dependency.	Detect, log, contain, and fail safely.
Workflow	Delay, duplicate, hide, or misroute the output.	Preserve correct user action, accountability, and escalation.

Appendix C - Minimum Public Reporting Dataset

Reporting category	Minimum public information
Identity	Title, authors, organization, document ID, version, date, status, funding, conflicts, and review status.
Intended use	Purpose, users, population, setting, modalities, output, and workflow role.
Data	Sources, dates, sites, devices, inclusion/exclusion, partitions, missingness, and preprocessing.
Reference standard	Definition, readers, blinding, adjudication, uncertainty, and limitations.
Model/system	Technology category and clinically relevant behavior without requiring proprietary source disclosure.
Testing	Internal/external design, sample size, metrics, uncertainty, calibration, subgroups, and robustness.
Human factors	Users, tasks, training, workflow, use errors, overrides, and alert behavior.
Utility	Comparator, endpoints, harms, process effects, and study design.
Lifecycle	Versioning, monitoring, change control, cybersecurity, and incident response.
Limitations	Known gaps, unsupported populations/settings, evidence boundaries, and unresolved research questions.

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xBxBio advances Connected Cardiovascular Intelligence™ through responsible integration of longitudinal cardiovascular information, multimodal clinical data, interoperability, clinical artificial intelligence, data integrity, provenance, governance, and human clinical oversight.

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